

Pandit Deendayal Energy University

End Semester Examination – May, 2025

B. Tech. Computer Science and Engineering

Semester - VI**Course Name: CRYPTOGRAPHY AND NETWORK SECURITY****Course Code: 23CP314T****Date: 13.05.2025****Time: 10:00 a.m. – 1:00 p.m.****Max. Marks: 100****Instructions:**

1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Q. No.	Description	Marks	CO's
1.	a) What is primitive root? Examine whether primitive roots exist modulo 26. Determine the number of primitive roots, if any exist.	5+5	CO3
	b) What is the key difference between a general group and an abelian group? Prove with example that the identity element in a group is unique.		
2.	a) Calculate the value of X modulo 17 with detail steps, where X is given by: $X \equiv 3^{140} + 5^{-1} \pmod{17}$	5+5	CO3
	b) Find the smallest positive integer x that satisfies the following congruence, and show your detailed steps: $4^x \equiv 13 \pmod{19}$.		
	-----Or-----		
2.	a) Use the Extended Euclidean Algorithm to find the smallest positive integer x such that: $35x \equiv 1 \pmod{48}$.	8+2	CO3
	b) Briefly explain what a Ciphertext-Only Attack (COA) is. What is the primary information available to the attacker when attempting a COA?		
3.	a) A number x satisfies the following system of congruences: $x \equiv 2 \pmod{3}$; $x \equiv 3 \pmod{5}$; $x \equiv 2 \pmod{7}$. Find the smallest positive x that satisfies these conditions using the Chinese Remainder Theorem.	10	CO3
	-----Or-----		
3.	a) List all the points generated by an elliptic curve group defined with the equation $E: y^2 = x^3 + 1 \pmod{7}$. Compute the product of $a = x^4 + x^3 + x^2 + 1$ and $b = x^3 + x^2 + x + 1$ in $GF(2^5)$ defined by irreducible polynomial $p(x) = x^5 + x^2 + 1$.	5+5	CO3
4.	a) Describe the three main algorithms (Key Generation, Encryption, and Decryption) that constitute the El-Gamal public-key cryptosystem. Explain the role of each step and how a recipient recovers the original plaintext from the ciphertext.	10	CO2
5.	a) Explain the fundamental concept of Identity-Based Encryption (IBE). How does IBE aim to simplify public key management compared to traditional Public Key Infrastructure (PKI)?	5+5	CO6

- b) Explain the core idea and purpose of Attribute-Based Encryption (ABE). How does ABE enable more flexible and fine-grained control over who can decrypt a ciphertext compared to a traditional public key encryption scheme
6. a) In a (3, 4) Shamir secret sharing scheme with modulus $p = 19$, three participants combine their shares: Participant 1: $(x_1, y_1) = (1, 10)$; Participant 2: $(x_2, y_2) = (2, 0)$; Participant 3: $(x_3, y_3) = (4, 11)$. Calculate the secret s that was shared. 10 CO3
7. a) Define network security and explain its primary objective. 3+3+4 CO1
 b) Explain how a firewall acts as a barrier between internal and external networks.
 c) What is an Intrusion Detection System (IDS)? Describe its objectives and mention two types of IDS.
8. a) In an organization, one compromised machine caused automatic spreading of malicious code to all computers on the same network without user intervention. Answer the following questions: 3+5+2 CO5
 I. What kind of malware does this represent? Explain your answer
 II. Explain how it differs from a traditional virus.
 b) How does a Trojan Horse typically enter a user's system?
9. a) A C-based login system uses `char user[10]; gets(user);` to accept usernames. When a user inputs more than 10 characters, the program crashes. 2+3+3+2 CO4
 I. Identify the vulnerability.
 II. How can this be exploited?
 III. Suggest two coding practices to prevent it.
 IV. Why is `gets()` unsafe?
10. a) A website allows users to log in by entering their username and password. The backend code uses the following SQL query to authenticate:
`SELECT * FROM users WHERE username = 'USER_INPUT' AND password = 'USER_PASSWORD';`
 An attacker enters ' OR '1' = '1' in the username field and leaves the password blank. This results in the query:
`SELECT * FROM users WHERE username = " OR '1' = '1' AND password = "`
 The condition '1' = '1' is always true, allowing the attacker to bypass authentication.
 I. What causes the SQL injection vulnerability in this case?
 II. What are the potential consequences of this attack?
 III. Suggest two methods to prevent SQL injection.
 b) Explain how SSL ensures confidentiality, integrity, and authentication in secure communication over the Internet.